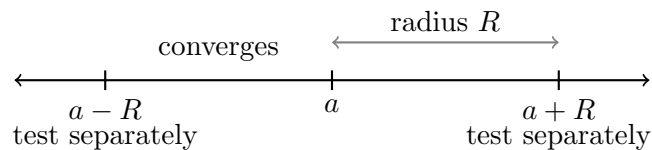


# Intervals of Convergence, Endpoint by Endpoint

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## Directions

- Every problem follows the same three moves: find the radius  $R$  from the ratio of coefficients, then test each endpoint by substituting it back into the original series, then write the interval with the right brackets.
- The ratio test never decides an endpoint — it returns 1 there by construction. Each endpoint needs its own test.
- Write intervals in interval notation. A square bracket means that endpoint converges; a round bracket means it does not.



How every problem on this page is laid out. No values shown: use the numbers from each series.

1. Consider  $\sum_{n=0}^{\infty} \frac{x^n}{n!}$ . The ratio of coefficients  $|c_n/c_{n+1}|$  simplifies to  $n + 1$ .

- Find the radius of convergence.
- Write the exact sum of the series at  $x = 2$ .
- State the interval of convergence and explain why no endpoint test is needed.

Answer: \_\_\_\_\_

**2.** Consider  $\sum_{n=1}^{\infty} \frac{x^n}{n 3^n}$ . The ratio of coefficients simplifies to  $\frac{3(n+1)}{n}$ .

- (a) Find the radius of convergence.
- (b) Write the exact sum of the series at  $x = 1$ .
- (c) Test both endpoints, naming the series each one produces, and state the interval of convergence.

Answer: \_\_\_\_\_

**3.** Consider  $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n^2}$ . The ratio of coefficients simplifies to  $\frac{(n+1)^2}{n^2}$ .

- (a) Find the radius of convergence.
- (b) Write the exact sum of the series at its right endpoint.
- (c) Test both endpoints and state the interval of convergence in interval notation.

Answer: \_\_\_\_\_

4. Consider  $\sum_{n=0}^{\infty} n! x^n$ . The ratio of coefficients simplifies to  $\frac{1}{n+1}$ .

- (a) Find the radius of convergence.
- (b) State the interval of convergence and explain what happens for every other value of  $x$ .

Answer: \_\_\_\_\_

5. Consider  $\sum_{n=1}^{\infty} \frac{(-1)^n (x+1)^n}{n 2^n}$ . The ratio of coefficients simplifies to  $\frac{2(n+1)}{n}$ .

- (a) Find the radius of convergence.
- (b) Write the exact sum of the series at its right endpoint,  $x = 1$ .
- (c) Test both endpoints and state the interval of convergence.

Answer: \_\_\_\_\_

6. Consider  $\sum_{n=1}^{\infty} \frac{x^n}{\sqrt{n}}$ . The ratio of coefficients simplifies to  $\frac{\sqrt{n+1}}{\sqrt{n}}$ .

(a) Find the radius of convergence.

(b) Find  $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}}$ , the size of the terms at either endpoint.

(c) The two endpoints give the same term sizes but different verdicts. Explain why, and state the interval of convergence.

Answer: \_\_\_\_\_

7. Consider  $\sum_{n=1}^{\infty} \frac{(3x-6)^n}{n}$ . Writing  $(3x-6)^n = 3^n(x-2)^n$  makes the coefficients  $c_n = 3^n/n$ , whose ratio simplifies to  $\frac{n+1}{3n}$ .

(a) Find the radius of convergence.

(b) Write the exact sum of the series at  $x = \frac{11}{5}$ .

(c) Test both endpoints, state the interval, and explain what goes wrong if the factor of 3 is not pulled out first.

Answer: \_\_\_\_\_

8. A dimensionless response index for a medication is modelled by  $S(d) = \sum_{n=1}^{\infty} \frac{(d-4)^n}{n 5^n}$ , where  $d$  is the dose in milligrams. The ratio of coefficients simplifies to  $\frac{5(n+1)}{n}$ .

- (a) Find the radius of convergence.
- (b) Write the exact value of  $S(6)$ , the index at a dose of 6 milligrams.
- (c) Test both endpoints, state the interval of convergence, and say which doses in milligrams the model is actually defined for.

Answer: \_\_\_\_\_